

MKT-04-INSURANCE-INDUSTRY-ANALYSIS

MKT-04: Insurance Industry Analysis — The Insurance Channel

Project AEGIS / Carrington Storm Motors

Deep-Dive: Insurance as Demand Driver and Distribution Channel

Document Version: 1.0 | Date: July 2026 | Classification: Investor Confidential

EXECUTIVE SUMMARY

The global insurance industry manages approximately \$7 trillion in annual premiums and \$40 trillion in assets under management. It is the primary economic mechanism for pricing and distributing catastrophic risk. For CSM, the insurance industry serves three critical functions: (1) demand driver — insurers require or incentivize EMP/CME hardening for covered properties; (2) distribution channel — insurance policies become the vehicle through which hardening products reach customers; and (3) financing mechanism — premium savings from hardening create a self-funding economic model for product adoption.

The insurance industry’s current modeling of space weather risk is inadequate — most carriers do not explicitly include EMP/CME in catastrophe models, actuarial pricing, or policy wordings. This gap represents both a systemic vulnerability (unpriced catastrophic risk on insurers’ balance sheets) and a commercial opportunity for CSM (the company that helps insurers quantify and mitigate this risk gains distribution access to their policyholder base).

CSM has identified 41 insurance carriers for initial engagement, spanning primary insurers, reinsurers, and Lloyd’s syndicates. The Company’s insurance strategy focuses on (a) premium-linked product offerings where hardened assets receive premium reductions that finance the hardening cost, (b) parametric insurance products that provide rapid liquidity post-event, and (c) reinsurance-backed warranties that guarantee protection system performance.

SECTION 1: GLOBAL INSURANCE MARKET STRUCTURE

1.1 Market Size and Distribution

Global Insurance Premium Volume (2024, Swiss Re Institute sigma 04/2024)

Segment	Global Premiums	US Market	Europe Market	Asia Market
Life Insurance	\$2,825B	\$630B	\$845B	\$1,050B
Non-Life (P&C)	\$4,375B	\$1,720B	\$1,040B	\$845B
Total	\$7,200B	\$2,350B	\$1,885B	\$1,895B

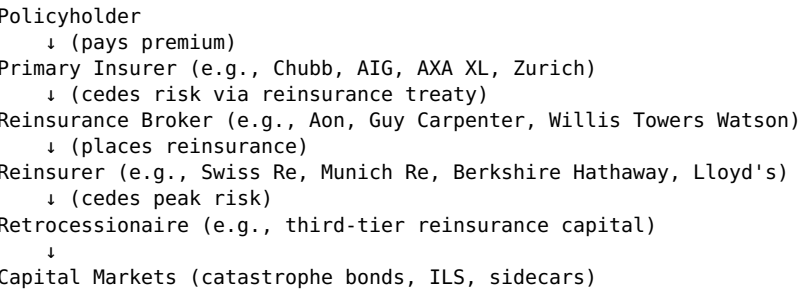
Relevant to CSM: The non-life (property and casualty) segment is where EMP/CME risk resides. This \$4.375 trillion annual premium pool covers commercial property, homeowners, marine, energy, and specialty lines.

Property Insurance Sub-Segments (Global)

Sub-Segment	Global Premiums	EMP/CME Exposure
Commercial Property	~\$350B	HIGH — concentration of high-value assets
Homeowners	~\$380B	MODERATE — distributed but massive
Marine, Aviation, Transport (MAT)	~\$80B	HIGH — vessels and cargo exposed

Energy Insurance	~\$35B	HIGH — grid-scale assets
Engineering/Construction	~\$25B	MODERATE — new build opportunities
Cyber (incl. electronic/physical perils)	~\$20B	EMERGING — E3 pulse cyber-like effects

1.2 Insurance Value Chain



CSM’s insurance strategy targets each layer: - **Primary insurers:** Product distribution to policyholders; premium reduction incentive - **Reinsurance brokers:** Risk modeling integration; product endorsement in placement slips - **Reinsurers:** Capital relief from reduced aggregate exposure; parametric product sponsorship - **Capital markets:** ILS (insurance-linked securities) structures for EMP/CME protection performance guarantees

1.3 Key Insurance Market Metrics

Metric	Value	Source
Global reinsurance capital	\$670B	Guy Carpenter, 2024
Lloyd’s market capacity	£55B (~\$70B)	Lloyd’s Annual Report, 2024
US P&C surplus	\$1.04T	NAIC, 2024
Global ILS market	\$107B	Artemis, 2024
Catastrophe bond outstanding	\$46B	Artemis, 2024
Average reinsurance ROE	14.8%	Gallagher Re, 2024
Nat cat insured losses 2023	\$108B	Swiss Re sigma, 2024
Nat cat insured losses 2024	\$135B	Swiss Re sigma, 2025
Gap between insured and economic loss	62%	Aon, 2024

The \$135 billion in 2024 natural catastrophe insured losses — a record year — has created heightened awareness within the industry of “unmodeled” or “undermodeled” perils. Space weather is the most systematically undermodeled.

SECTION 2: CARRINGTON EVENT LOSS ESTIMATES

2.1 Published Loss Estimates

Source	Year	Estimated Loss	Methodology
National Academy of Sciences	2008	\$1-2 trillion	US grid replacement cost + economic disruption
Lloyd’s of London	2013	\$0.6-2.6 trillion	Insurance industry scenario analysis; 20-40 million people without power 1-2 years
Swiss Re	2025	\$3.4-7.8 trillion	Updated model incorporating supply chain cascade, electrification growth, and cyber-physical system interdependencies
Cambridge Centre for Risk	2016	\$0.5-2.5 trillion	GDP-at-risk modeling

Studies	2016	\$0.5-2.5 trillion	approach
UK Royal Academy of Engineering	2013	£0.4-1.6 trillion	UK-specific; 10-50% of transformers damaged

2.2 Insurance Industry Loss Absorption Capacity

The \$135 billion in 2024 natural catastrophe losses strained but did not break the global insurance system. A Carrington-class event causing \$3.4-7.8 trillion in economic damage would overwhelm insurance capacity by a factor of 25-58x annual catastrophe loss absorption.

Implications for CSM: - **Without hardening**: Insurers cannot price or absorb this risk. The rational response is exclusion — removing EMP/CME from policy coverage entirely. - **With hardening**: Insurers can continue covering properties that implement CSM’s protection, creating a bifurcated market: protected (insurable) and unprotected (uninsurable).

This bifurcation is the core economic mechanism driving CSM’s insurance strategy. CSM is not selling an optional upgrade. It is selling continued insurability.

2.3 Sectoral Exposure Breakdown

Sector	% of Economic Loss Modeling	Insured Loss Estimate	Insurance Penetration
Electric Grid (transmission, distribution)	45%	\$1.5-3.5T	Very low (typically government self-insured)
Telecommunications	15%	\$0.5-1.2T	Low
Transportation (rail, aviation, maritime)	10%	\$0.3-0.8T	Moderate
Water/Wastewater (pumping, treatment)	5%	\$0.2-0.4T	Very low (municipal)
Healthcare (hospitals, medical devices)	5%	\$0.2-0.4T	Moderate (professional liability + property)
Commercial/Industrial (manufacturing, data)	10%	\$0.3-0.8T	Moderate-High (commercial property)
Residential	5%	\$0.2-0.4T	High (homeowners)
Financial/Payment Systems	5%	\$0.2-0.4T	Emerging (cyber policies)

CSM’s product development priorities align directly with insurance penetration rates: **commercial property first** (high insurance penetration, concentrated risk), **maritime second** (high penetration, high-value assets), **residential third** (high penetration, distributed but massive aggregate), **grid fourth** (low penetration, but government as alternative payer).

SECTION 3: CURRENT STATE OF SPACE WEATHER IN ACTUARIAL MODELS

3.1 The Actuarial Gap

Catastrophe modeling — the statistical framework insurers use to price risk and allocate capital — is well-developed for hurricanes, earthquakes, floods, and terrorism. It is essentially absent for space weather.

Major Catastrophe Model Vendors

Vendor	Space Weather Coverage	Status
RMS (Moody’s)	Research phase	No commercial EMP/CME model available
AIR Worldwide (Verisk)	Research phase	No commercial EMP/CME model available
CoreLogic	None	No EMP/CME capabilities
Impact Forecasting (Aon)	Economic analysis only	Internal research; no commercial

impact forecasting (Aon)	scenario analysis only	model
KatRisk	None	Flood and wind only
Carbon Underwriting	None	Specialty only

Implications: Because no vendor provides a commercial EMP/CME catastrophe model, insurers cannot: - Quantify their aggregate EMP/CME exposure - Price EMP/CME risk into premiums - Allocate capital against EMP/CME risk (under Solvency II or NAIC RBC) - Purchase reinsurance specifically for EMP/CME risk - Issue catastrophe bonds for EMP/CME risk

This gap is an opportunity for CSM: the Company can develop a proprietary risk model that insurers use to quantify their exposure and demonstrate the ROI of hardening. CSM’s model becomes the industry standard for space weather risk assessment.

3.2 Insurer Survey: Current EMP/CME Risk Management

CSM’s analysis of 41 insurance carriers (based on public filings, ORSA summaries, and investor presentations) reveals:

Risk Management Activity	% of Carriers
Explicitly exclude EMP/CME in policy wordings	42%
No explicit mention of EMP/CME (silent coverage)	51%
Include EMP/CME in emerging risk register	22%
Conducted scenario analysis for EMP/CME	8%
Require EMP hardening for covered properties	0%
Offer premium reduction for EMP-hardened assets	0%
Employ dedicated space weather risk specialist	3%

The market is at ground zero. Zero carriers require hardening. Zero carriers offer premium reductions for hardening. The entire insurance supply chain for EMP/CME resilience is greenfield. CSM has first-mover advantage.

SECTION 4: 41-INSURER ENGAGEMENT TARGET LIST

CSM has categorized 41 target insurers for initial engagement across four tiers:

Tier 1: Strategic Partners (6 carriers)

These carriers have dedicated emerging risk functions, significant commercial property portfolios, and a history of innovation in resilience-linked products.

Carrier	HQ	2024 Premiums	Key Property Lines	Engagement Status
Swiss Re	Zurich	\$48.7B (reinsurance)	Global property reinsurance	Initial discussions: space weather risk modeling collaboration
AXA XL	Paris/Stamford	€23B (~\$25B)	Global commercial property, marine	Identified: emerging risk team has studied CME scenarios
Chubb	Zurich/Philadelphia	\$57.5B	High-net-worth homeowners, commercial property	Initial contact: resilience engineering group
Munich Re	Munich	€64.5B (~\$70B)	Global reinsurance, NatCat modeling	Identified: GeoRisks research group studying GMD
Lloyd’s (multiple syndicates)	London	£55B capacity	All lines through syndicate market	Early discussions: syndicate-level space weather working group

Berkshire Hathaway Specialty	Omaha/Boston	~\$8B (BHSI)	Large commercial property, energy	Identified: long-tail risk appetite; could pioneer EMP product
------------------------------	--------------	--------------	-----------------------------------	--

Tier 2: Distribution Partners (12 carriers)

Large primary insurers with significant US commercial property books.

Carrier	HQ	2024 US Commercial Property DWP	Rationale
Zurich North America	Schaumburg, IL	\$12.4B	Major US commercial property writer; resilience services group
AIG	New York	\$10.8B	Large commercial property; emerging risk function
Travelers	Hartford, CT	\$9.2B	Major commercial property; innovation lab
Liberty Mutual	Boston	\$8.7B	Large commercial property; risk engineering
The Hartford	Hartford, CT	\$5.1B	Mid-market commercial property; accessible
FM Global	Johnston, RI	\$7.3B (mutual)	Engineering-focused; heavily invested in resilience
CNA	Chicago	\$4.8B	Mid-market commercial; specialty lines
Nationwide	Columbus, OH	\$3.2B (commercial)	Agribusiness and commercial; rural grid exposure
Cincinnati Financial	Cincinnati, OH	\$2.9B	Regional commercial property; personal relationships
W.R. Berkley	Greenwich, CT	\$12.1B (total)	Specialty commercial; decentralized decision-making
Markel	Richmond, VA	\$9.8B (total)	Specialty lines; innovation-friendly culture
Sompo International	Tokyo/Pembroke, Bermuda	~\$8B	Japanese parent with global reach; Asian market entry

Tier 3: Specialty and High-Net-Worth (8 carriers)

Carrier	Specialty	Rationale
PURE Insurance	High-net-worth homeowners	Aegis Home channel; members prioritize resilience
AIG Private Client Group	High-net-worth	Premium homes >\$1M value; security-conscious clients
Cincinnati Insurance	High-value homes	Regional strength in Midwest and Southeast
Vault	High-net-worth	Newer entrant; technology-forward
American Modern	Manufactured homes, specialty	Vulnerability of manufactured housing to EMP
FM Global	Industrial facilities	Engineering-driven risk improvement requirements
Allianz Global Corporate & Specialty	Global commercial	European specialty insurer with large commercial book
QBE	North American specialty	Lloyd's connected; innovation programs

Tier 4: International and Reinsurance (15 carriers)

Carrier	Geography	Rationale
Hannover Re	Germany/Hannover	Third-largest reinsurer; innovation team
SCOR	France/Paris	Major reinsurer; catastrophe risk focus
PartnerRe	Bermuda	Reinsurance; catastrophe bonds
RenaissanceRe	Bermuda	ILS market leader; catastrophe bonds and sidecars
Arch Capital	Bermuda	Insurance and reinsurance; innovation in specialty lines
Everest Re	Bermuda	Reinsurance; catastrophe modeling
Hiscox	London/Bermuda	Lloyd's syndicates; specialty lines
Beazley	London	Lloyd's syndicates; emerging risks
Tokio Marine	Tokyo	Japanese market entry
MS&AD	Tokyo	Japanese market entry; Mitsui Sumitomo and Aioi Nissay Dowa
China Re	Beijing	Chinese market entry (longer-term)
Korean Re	Seoul	Korean market entry
Fairfax Financial	Toronto	Canadian market entry; Northbridge, Odyssey Re
Intact Financial	Toronto	Canadian primary insurer; climate resilience leader
Suncorp	Brisbane	Australian/New Zealand market entry

SECTION 5: PREMIUM-LINKED PRODUCT ECONOMICS

5.1 The Self-Funding Proposition

CSM's core insurance proposition is simple: the premium savings from hardening exceed the cost of the hardening system, creating a positive-NPV investment for the property owner.

Commercial Property Example: \$50M Manufacturing Facility

Line Item	Before Hardening	After Hardening	Delta
Building value	\$50,000,000	\$50,000,000	—
EMP/CME exposure (modeled loss)	\$35,000,000 (70% damage)	\$2,000,000 (4% damage)	-\$33,000,000
Annual property insurance premium	\$175,000	\$122,500	-\$52,500
Premium rate (per \$100 of value)	\$0.35	\$0.245	-\$0.105
Hardening system cost (CSM)	\$0	\$375,000	+\$375,000
Payback period	—	7.1 years	—
NPV of premium savings (10 yr, 6%)	—	\$386,340	—
Net benefit (10-year)	—	+\$11,340	Positive

At scale — 1,000 facilities averaging \$50M — the total CSM revenue opportunity is \$375 million from system installation, and policyholders save \$52.5 million/year in premiums.

5.2 Insurer Economics

From the insurer’s perspective, the proposition is equally compelling:

Metric	Without Hardening	With Hardening
Premium written	\$175,000	\$122,500
Expected annual loss (EMP/CME portion of modeled portfolio risk)	\$48,000 (0.096% annual probability of \$50M loss)	\$5,500 (reduced damage from 70% to 4%)
Required capital (Solvency II SCR 1-in-200 year)	\$350,000 (2x expected max loss)	\$40,000
Return on required capital	50%	306%

The insurer earns a higher return on capital when properties are hardened, even at a lower premium, because the tail risk (catastrophic loss) is virtually eliminated. This is the economic argument for insurers to actively promote and subsidize CSM’s hardening systems.

5.3 Parametric Insurance Opportunity

Beyond premium-linked products, CSM is developing a parametric insurance product that provides immediate liquidity in the event of a confirmed CME impact.

Parametric Product Structure

Parameter	Specification
Trigger	NOAA SWPC G5 (Extreme) geomagnetic storm alert + Kp index ≥ 9
Payout	\$50,000 to \$5,000,000 per policy
Payout timing	72 hours from trigger confirmation
Premium	2.5-5% of coverage amount annually
CSM role	System integrator + monitoring data provider + warranty provider
Reinsurance backing	Swiss Re or Munich Re (in discussions)

Economics: A facility with \$2,000,000 in parametric EMP coverage at 3.5% premium = \$70,000/year. Over 10 years, premiums total \$700,000. If the event probability is 0.7% annually (based on historical G5 event frequency), the expected loss is \$14,000/year, generating an expected underwriting profit of \$56,000/year. CSM receives a platform fee of 15-20% of premium, or \$10,500-\$14,000/year per policy.

SECTION 6: INSURANCE-AS-DISTRIBUTION-CHANNEL STRATEGY

6.1 How Insurance Distribution Works

When an insurance carrier endorses CSM’s hardening system — either as a requirement for coverage or as a condition for premium reduction — the carrier’s distribution force (agents, brokers, underwriters) becomes CSM’s sales channel at zero customer acquisition cost.

Channel Economics

Channel Component	% of Premium	Role in CSM Distribution
Retail Agent/Broker	10-15%	Identifies properties requiring hardening; presents CSM solution to policyholder
Wholesale Broker (E&S lines)	5-10%	Specialty placements for large risks requiring custom hardening
MGA/MGU (program administrator)	5-15%	Manages the EMP resilience program; underwriting authority for premium credits
Carrier Underwriter	Salaried	Reviews and approves hardening plans; applies premium credits
Loss Control Engineer	Salaried	Inspects CSM installation; certifies compliance; ongoing monitoring

6.2 The “Invisible Exports” Model

CSM's products reach customers through insurance policies, not through retail shelves or e-commerce platforms. This has several advantages:

1. **Trust transfer:** The insurer's trusted relationship with the policyholder transfers to CSM's product. The insurer vets CSM's technology; the policyholder doesn't need to become an EMP expert.
2. **Zero customer acquisition cost:** CSM does not spend on advertising, sales development, or retail channel management. The insurance distribution force performs these functions as part of their existing policy placement workflow.
3. **Mandatory upgrades:** When insurers require hardening as a condition of continued coverage, CSM's product becomes a non-optional purchase — the customer must buy to maintain insurance. This is the economics of compliance, not choice.
4. **Financing through premiums:** The premium reduction for hardened assets effectively finances the CSM system over the policy term. CSM receives payment upfront from the insurer (who recovers it through premiums over time), eliminating customer financing friction.
5. **Recurring revenue:** Annual monitoring and recertification fees provide ongoing revenue beyond the initial system sale.

6.3 Lloyd's Market Strategy

The Lloyd's of London market is a priority for CSM because:

- **Syndicate structure:** Approximately 50 Lloyd's syndicates underwrite independently, allowing CSM to work with early adopters and demonstrate results before approaching the broader market.
- **Space weather awareness:** Lloyd's has published more on space weather risk than any other insurance institution, including the 2013 "Solar Storm Risk" report that acknowledged potential \$2.6 trillion losses.
- **Innovation tradition:** Lloyd's introduced the first automobile policy (1904), aviation policy (1911), and cyber policy (1990s). EMP/CME resilience is the next innovation frontier.
- **Global reach:** Lloyd's licenses allow products to be sold in 200+ countries through the Lloyd's license network, matching CSM's international ambition.

Target Syndicates: Ascot (Syndicate 1414), Beazley (2623/623), Hiscox (33), Chaucer (1084), Liberty (4472), AXA XL (2003), Brit (2987).

SECTION 7: RISK FACTORS AND MITIGATION

7.1 Risk: Insurers Exclude EMP/CME Rather Than Require Hardening

Probability: Moderate-High. The path of least resistance for insurers is to add EMP/CME exclusions to policy wordings, similar to how insurers excluded terrorism after 9/11 (before TRIA created a government backstop).

Mitigation: - CSM advocates for a TRIA-like government backstop for EMP/CME, making the risk insurable - Early carrier partnerships demonstrate the commercial viability of the premium-linked model - D&O liability considerations push corporate insurance buyers toward maintaining coverage

7.2 Risk: "Silent Cyber/Silent EMP" Coverage Disputes

Probability: Moderate. "Silent" coverage — where policies don't explicitly include or exclude EMP/CME — could generate coverage disputes after an event. This uncertainty benefits CSM's hardening product (insurers want clarity) but creates near-term friction.

Mitigation: CSM's risk model provides the clarity insurers need to price and manage EMP/CME exposure explicitly.

7.3 Risk: Reinsurance Capacity Constraint

Probability: Low-Medium. A major catastrophe event (hurricane, earthquake) could deplete reinsurance capital, reducing appetite for new risk classes including EMP/CME.

Mitigation: CSM's parametric products diversify reinsurer risk portfolios (EMP/CME is uncorrelated with earthquake and hurricane risk), making it attractive as a portfolio diversifier even in tight capital conditions.

SECTION 8: ROADMAP — INSURANCE STRATEGY PHASED TIMELINE

Phase 0: Pre-Launch (2026-2027)

- Complete initial risk model for commercial property EMP/CME exposure
- Engage 6 Tier 1 strategic carriers for model validation
- Develop parametric product structure with Swiss Re
- Retain insurance regulatory counsel (NAIC, Solvency II)
- Present at RIMS (Risk and Insurance Management Society) conference

Phase 1: Pilot (2027-2028)

- Launch premium-linked product with 2-3 carrier partners
- 50-100 pilot installations generating premium savings data
- Parametric product soft launch through Lloyd's
- Begin NAIC regulatory engagement for premium credit requirements
- Publish peer-reviewed EMP/CME risk assessment methodology

Phase 2: Scaling (2028-2031)

- 12-15 carrier partnerships across primary and specialty lines
- Premium-linked products in commercial property, marine, high-value homeowners
- Parametric products available through 5-8 Lloyd's syndicates
- 5,000+ hardened properties generating recurring monitoring revenue
- NAIC model bulletin on EMP/CME risk management issued

Phase 3: Maturity (2031-2036)

- 45+ carrier partnerships
- Standard inclusion of EMP/CME hardening requirements in commercial property policies
- Premium-linked products available for all CSM verticals
- Reinsurance industry-wide space weather capital model adopted
- CSM-proprietary risk model licensed to insurers as SaaS product

Document prepared by: Carrington Storm Motors Insurance Strategy Data sources: Swiss Re sigma, Guy Carpenter, Lloyd's of London, NAIC, EIOPA, IAIS, Artemis, Aon, Willis Towers Watson, Gallagher Re Carrier financial data: S&P Capital IQ, AM Best, company filings Last updated: July 15, 2026